Hyeonjun An

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RESEARCH PROFILE

Experimental soft-matter physics using quantitative neutron and synchrotron X-ray imaging: polymer swelling and drying, water-vapor uptake by hygroscopic liquids, droplet freezing, and transport in porous materials.

EDUCATION

Combined M.S. and Ph.D. in Physics, UNIST Feb 2022 – Present

Expected graduation: February 2027 | Advisor: Prof. Joonwoo Jeong

B.S. in Physics and Energy Engineering, UNIST Mar 2016 – Feb 2022

Double major | Summa cum laude

Student exchange, Hong Kong University of Science and Technology (HKUST), Summer 2017.

RESEARCH EXPERIENCE

Doctoral Researcher, Soft Matter Physics Laboratory, UNIST 2022 – Present

- Quantify swelling and drying in hydrogels and PDMS using H/D isotope contrast, inverse Abel reconstruction, and poromechanical models.
- Map water-vapor uptake and concentration gradients in hygroscopic droplets and capillaries; investigate gravity, diffusion and mixing.
- Resolve freezing fronts, polymer redistribution and brine networks using synchrotron radiography, tomography and Raman mapping.
- Measure liquid distributions and wetting fronts during imbibition of evaporative liquids in heterogeneous porous materials.

PAL research placement (Apr–Dec 2024): 50 days at PLS-II 6C BMI for synchrotron X-ray imaging.

Research internships: KRISS, Multiscale Length Metrology Team (Jan–Feb 2022); UNIST, Soft Matter Physics Laboratory (Mar–Dec 2021).

PUBLICATIONS AND MANUSCRIPTS

1. Hyeonjun An, B. Kim, J. K. Im, S. Kim, M. W. Kim, S.-G. Kim, J.-H. Lim, K. Kim, J. Jeong. **Polymer-Regulated Freezing of Water Droplets Revealed by Synchrotron X-ray Imaging and Raman Spectroscopy.** *Journal of Colloid and Interface Science*, Article 141826 (2026), advance online publication. DOI: 10.1016/j.jcis.2026.141826.
2. Hyeonjun An, Y. Koo, J. Cheon, P. Trtik, J. Jeong. **Condensation of the Vapor in Sessile Water Droplets and Open-Ended Water-Filled Capillaries Revealed by Neutron Radiography.** *Springer Proceedings in Physics* 348, 235–243 (2026). DOI: 10.1007/978-3-032-15003-5_27.
3. H. Yang, G. Ji, M. Choi, S. Park, Hyeonjun An, et al. **Suppressed terahertz dynamics of water confined in nanometer gaps.** *Science Advances* 10, eadm7315 (2024). DOI: 10.1126/sciadv.adm7315.
4. J. K. Im, Hyeonjun An, S.-G. Kim, J.-H. Lim, J. Jeong. **Direct X-ray visualization of bubble-mediated singularity formation in freezing droplets.** *Newton*, accepted (2026). arXiv:2511.20571.

AWARDS AND SCHOLARSHIPS

Outstanding Poster Presentation Award Oct 2026
Korea Physical Society Fall Meeting.

First Prize for polymer swelling poster Oct 2026
10th International Topical Meeting on Neutron Radiography (ITMNR-10).

Outstanding Presentation Award in Oral Session 2025; 2026
Korean Physical Society Spring Meetings.

Excellence Award, DRB Space Challenge Aug 2026
Drop-tower microgravity experiment.

Misan Mountaineering Scholarship, UNIST (Fall 2026).

UNIST graduate and teaching assistant scholarships (2021–2026).

National Science and Engineering Undergraduate Scholarship, Korea Student Aid Foundation (2016–2018; 2021).

RESEARCH FUNDING

Principal Investigator, KONICOF Aug 2026 – Feb 2027
2026 Joint Research Program for Ph.D. Students Utilizing Overseas Research Facilities.

Quantitative neutron imaging of mass transport in soft matter with complementary X-ray imaging.

Korea Nuclear International Cooperation Foundation | International partner: Paul Scherrer Institute, Switzerland.

EXPERIMENTAL AND ANALYTICAL EXPERTISE

Neutron radiography: 70 beamtime days at PSI ICON and NEUTRA, Switzerland, and HANARO ENF, Korea.

Synchrotron X-ray imaging: 15 beamtime days, plus a separate 50-day PAL research placement. Beamlines: PAL PLS-II 6C BMI, 9D and 7C, Korea.

Synchrotron scattering and diffraction: PAL PLS-II 6D and 9A, Korea.

Analysis and modeling: H/D isotope contrast, inverse Abel reconstruction, tomography and segmentation; Flory–Rehner/Darcy and Maxwell–Stefan models.

SELECTED CONFERENCE PRESENTATIONS

Korea Physical Society Fall Meeting Oct 2026
Poster: “Connectivity of Brine Networks and Tip Morphology in Freezing Saline Droplets.”

10th International Topical Meeting on Neutron Radiography (ITMNR-10) Oct 2026
Two posters: “Quantifying Polymer Swelling Kinetics via Neutron Radiography” and “Direct Observation of Water Vapor Uptake in Hygroscopic Liquids by Neutron Radiography.”

Korean Physical Society Spring Meeting Apr 2026
Oral: “Quantifying Polymer Swelling Kinetics via Neutron Radiography.”

Korean Physical Society Spring Meeting Apr 2025
Oral: “Direct Observation of Water Vapor Uptake in Hygroscopic Liquids by Neutron Radiography.”

TEACHING AND SERVICE

Teaching Assistant, UNIST (2018; 2022–2026): General Physics I and II, General Physics Laboratory I, Mathematical Physics, and Thermal and Statistical Physics.

Military service (2019–2021): Motor transport operator, 8th Fighter Wing, Republic of Korea Air Force.